

Day 4: Tasks

1. **Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.**



RAM Reseating Procedure

Steps Performed

1. Shut down the computer completely.
 2. Unplugged the power cable and disconnected all peripherals.
 3. Pressed the power button for a few seconds to discharge any remaining power.
 4. Opened the computer case/laptop back cover using a screwdriver.
 5. Located the RAM module installed in the motherboard's RAM slot.
 6. Released the retaining clips on both sides of the RAM slot.
 7. Carefully removed the RAM module.
 8. Inspected the RAM and slot for dust or visible damage.
 9. Reinserted the RAM module into the same slot, ensuring proper alignment with the notch.
 10. Pressed firmly until the retaining clips locked into place.
 11. Reassembled the computer and reconnected the power cable.
 12. Powered on the system and verified that it booted normally.
2. **Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.**
Hint: Think about display issues, driver checks, and physical connections.



GPU Troubleshooting Checklist

Step	Action
1	Identify the symptoms (no display, screen flickering, artifacts, game crashes, or low FPS).
2	Check that the monitor cable is securely connected to the GPU and that the monitor is powered on.
3	Verify that the GPU is properly seated in the PCIe slot and that all required PCIe power connectors are attached.
4	Check Device Manager for GPU detection and look for any warning symbols or error codes.
5	Update, reinstall, or roll back the graphics drivers if the issue started after a driver update.
6	Monitor GPU temperatures using software such as HWMonitor or MSI Afterburner to check for overheating.
7	Test the GPU in another PC, or test another known-good GPU in the system to isolate the fault.
8	Restart the system and run a game or benchmark to confirm that the display and performance issues are resolved.

3. **Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.**



Storage Device Troubleshooting Simulation

Before Reconnecting the Drive

1. I shut down the computer completely.
2. I disconnected the SSD/HDD from the system.
3. After turning the computer back on, I entered the BIOS setup.

Observations:

- The computer was unable to find a bootable device.
- An error such as "**No Bootable Device Found**" or "**Operating System Not Found**" appeared during startup.
- The SSD/HDD was missing from the BIOS storage device list.
- The drive was also absent from the boot priority menu.

After Reconnecting the Drive

1. I turned off the computer again.
2. I reconnected the SSD/HDD and made sure all connections were secure.
3. I restarted the system and checked the BIOS.

Observations:

- The SSD/HDD was detected successfully in the BIOS.
- The drive appeared in the boot priority list.
- The computer booted normally into Windows without any error messages.
- All files and applications were accessible as expected.

4. **Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.**



Component to Check First: RAM

Reason:

RAM issues are a common cause of random restarts, system freezes, crashes, and blue screen errors. If the RAM is faulty, loose, or not functioning correctly, the computer may restart unexpectedly while running programs or even during normal use.

Diagnostic Steps

1. Shut down the computer and disconnect the power.
2. Open the case and check that the RAM modules are properly seated in their slots.
3. Remove and reinsert the RAM modules securely.
4. If there are multiple RAM sticks, test the system with one stick at a time.
5. Run **Windows Memory Diagnostic** or a memory testing tool to check for errors.
6. Monitor the system for random restarts after testing each RAM module.
7. If errors are found, replace the faulty RAM stick and test again.